



Intergenic lncRNA
Intronic lncRNA
Antisense lncRNA
Sense lncRNA
Enhancer lncRNA
Bidirectional lncRNA
...

Identification and Characterization of lncRNAs from Bulk RNA-seq Data

From Raw Reads to Functional Non-Coding RNA Insights

RNA-seq Workshop | Bioinformatics & Computational Biology

*Non-coding RNAs
Big Roles
in Biology*

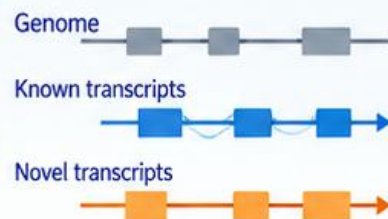
Regulation
Epigenetics
Development
Disease
Precision Medicine

Raw RNA-seq Reads



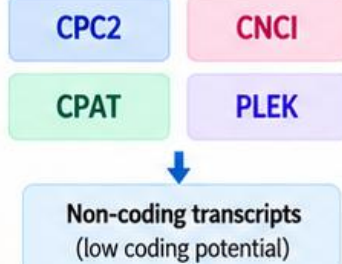
- Raw reads (FASTQ)
- Quality control (e.g., FastQC)
- Read trimming (e.g., Trimmomatic)
- Remove low-quality reads

Transcript Discovery



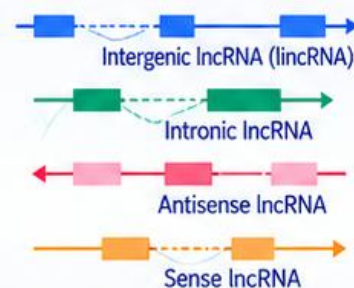
- Genome-guided / de novo transcript assembly (e.g., StringTie, HISAT2)
- Obtain known and novel transcripts

Coding-Potential Analysis



- Assess coding potential
- Remove protein-coding transcripts

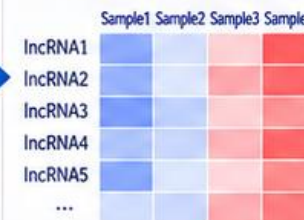
lncRNA Identification



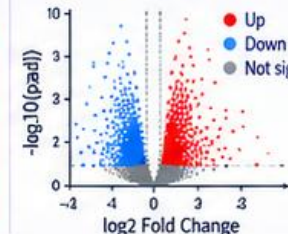
- Apply filters (length, exon number, etc.)
- Classify lncRNAs (lincRNA, intronic, antisense, sense, etc.)
- Obtain high-confidence lncRNAs

Quantification & Differential Expression

Transcript quantification
(StringTie / featureCounts)



Differential expression
(DESeq2)



- Generate expression matrix (lncRNAs)
- Identify differentially expressed lncRNAs
- Up- and down-regulated lncRNAs ($p_{adj} < 0.05$, $|\log_2FC| \geq 1$)

Functional Analysis

GO Enrichment



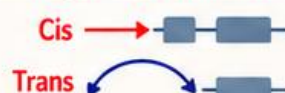
- Biological process
- Molecular function
- Cellular component

KEGG Pathway Analysis



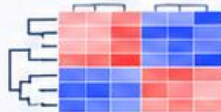
- Signaling pathways
- Metabolic pathways
- Disease pathways

Target Gene Prediction



- Cis-acting targets (nearby genes)
- Trans-acting targets (co-expression, correlation)

Co-expression Analysis



- lncRNA-mRNA correlation
- Co-expression modules (e.g., WGCNA)

PPI / Network Analysis



- lncRNA-mRNA networks
- Hub genes/modules
- Cytoscape visualization

Disease Association



- lncRNA-disease links (e.g., LncRNADisease)
- Literature and database annotation

Biological Interpretation



- Key lncRNAs and pathways
- Potential regulatory mechanisms
- Insights into biological functions and disease relevance



What is RNA?

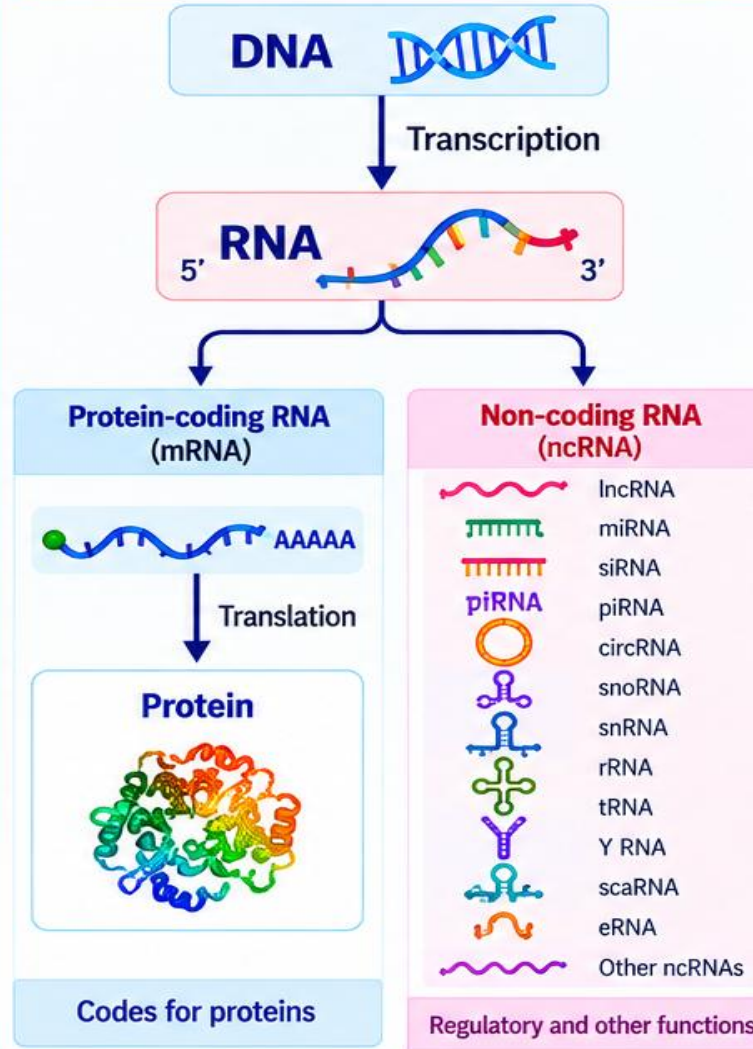
A diverse family of molecules with multiple functions in the cell

Beyond mRNA
Many RNAs, Many Roles, One Life

RNA molecules can broadly include:

mRNA		Messenger RNA (codes for proteins)	~ 0.5–10 kb
lncRNA		Long non-coding RNA (regulatory functions)	> 200 nt
miRNA		MicroRNA (post-transcription)	~ 20–24 nt
siRNA		Small interfering RNA (gene silencing)	~ 20–25 nt
piRNA		PIWI-interacting RNA (transposon silencing)	~ 24–32 nt
circRNA		Circular RNA (covalently closed loop)	~ 0.1–10 kb
snoRNA		Small nucleolar RNA (rRNA modification)	~ 60–300 nt
snRNA		Small nuclear RNA (splicing)	~ 100–300 nt
rRNA		Ribosomal RNA (ribosome component)	~ 120–5,000 nt
tRNA		Transfer RNA (amino acid transport)	~ 73–93 nt
Y RNA		Y RNA (DNA replication, Ro RNP)	~ 80–120 nt
scaRNA		Small Cajal body RNA (snRNA modification)	~ 70–200 nt
eRNA		Enhancer RNA (enhancer activity)	~ 0.1–2 kb
Other ncRNAs		e.g., vault RNA, 7SK RNA, telesRNA, etc.	Variable

Central concept



Key takeaways

- 1 RNA is a diverse group of molecules.
- 2 Both coding and non-coding RNAs are transcribed from DNA.
- 3 Non-coding RNAs (e.g., lncRNA, miRNA, circRNA, etc.) have important regulatory and functional roles.
- 4 Different RNA types vary in size, structure and cellular localization.
- 5 RNA molecules work together to control gene expression, genome stability and cellular processes.



Major functions of RNAs

Protein-coding RNA

- Carry genetic information
- Code for proteins

Non-coding RNAs

- Regulate gene expression
- RNA processing (splicing, modification)
- Chromatin regulation
- Genome stability
- Cellular structure and function
- Involved in development, disease and environmental responses



Take-home message

RNA is not just mRNA.
A diverse and complex RNA world plays essential roles in life.

Different RNAs Drive Biology!



What is a lncRNA?

Long transcripts. No protein. Big functions.

*Non-coding RNA
for a complex world*

Definition

Long non-coding RNA (lncRNA) refers to a transcript that:

- 1 Is longer than 200 nucleotides
- 2 Has little or no protein-coding potential
- 3 Is transcribed from the genome
- 4 Is often tissue-, cell- or condition-specific

General features of lncRNAs



Length
> 200 nt



Non-coding
No or very low
protein-coding
potential



**Transcribed
from genome**
by RNA polymerase II



**Tissue-/cell-
specific**
Dynamic and
context-dependent

lncRNA vs mRNA

Feature	lncRNA	mRNA
Length	> 200 nt	Usually > 200 nt
Protein-coding potential	Little or none	High
Open reading frame (ORF)	Short / no ORF	Long ORF
Expression level	Generally lower	Generally higher
Sequence conservation	Lower	Higher
Function	Regulatory roles	Codes for proteins

Major categories of lncRNAs

Type	Description	Schematic representation
Intergenic (lincRNA)	Transcribed from regions between protein-coding genes	
Antisense	Transcribed from the opposite strand of another gene	
Intronic	Derived from introns of protein-coding genes	
Sense-overlapping	Overlaps with a protein-coding gene on the same strand	
Bidirectional	Transcribed near another gene (usually within 1 kb, opposite direction)	

Functions of lncRNAs



**Chromatin
regulation**

Recruit chromatin
modifiers, alter
epigenetic state



**Transcriptional
regulation**

Enhance or repress
gene expression



**Post-transcriptional
regulation**

Affect splicing,
stability and
translation



Scaffold / decoy

Interact with proteins,
RNAs or DNA



Key message

**lncRNAs are not “transcriptional noise” —
they are important regulators of gene expression
and cellular processes.**

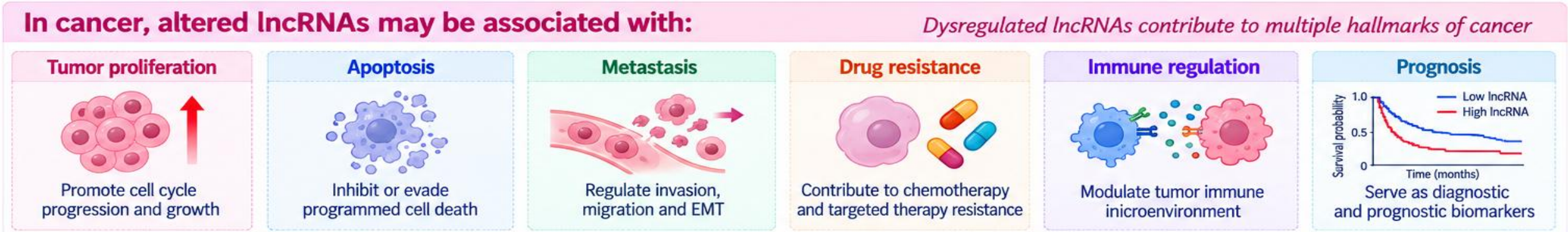
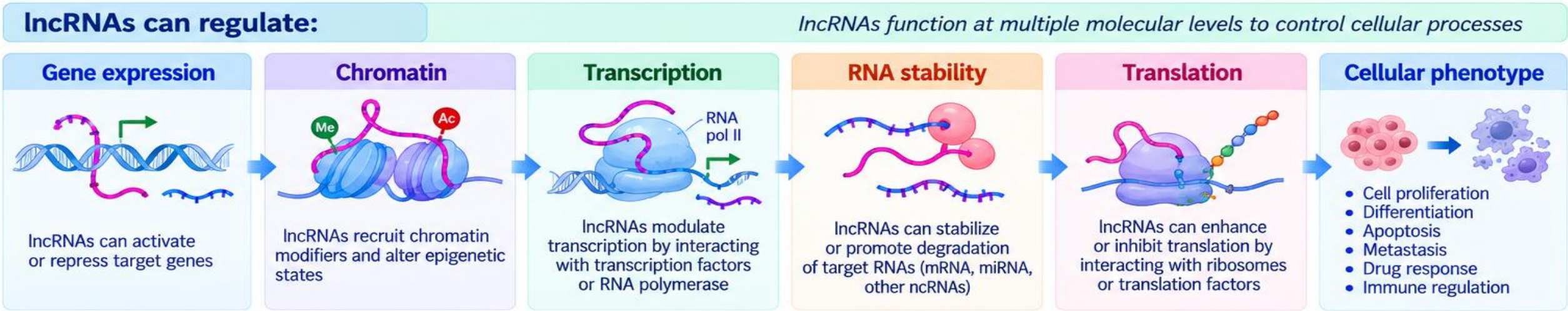
- ✓ Diverse in origin and function
- ✓ Dynamic and context-specific expression
- ✓ Involved in development, physiology and disease

*Small RNAs.
Big Impact!*



lncRNAs: Expanding the functional landscape of the genome

*Non-coding RNAs
Regulate the future*





Key takeaways

- ✓ lncRNAs are important regulators of gene expression and cellular processes.
- ✓ Dysregulated lncRNAs play critical roles in cancer development and progression.
- ✓ lncRNAs have potential as diagnostic, prognostic and therapeutic targets.



Overall message

Studying lncRNAs helps us understand disease mechanisms and opens new opportunities for biomarker discovery and therapeutic intervention.

lncRNAs Bridge the Gap from Genes to Disease!

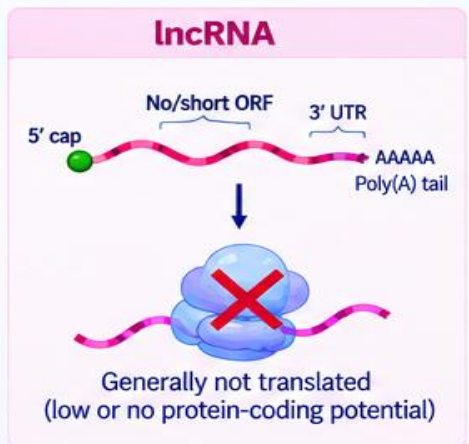
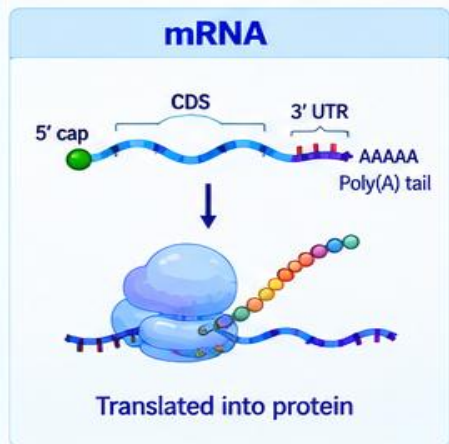
Different
Transcripts
Different Functions

lncRNA vs mRNA

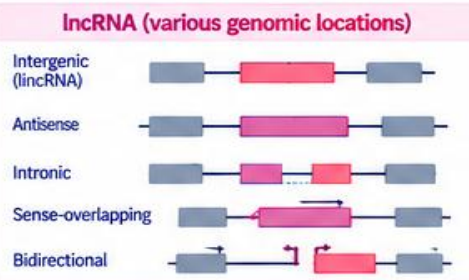
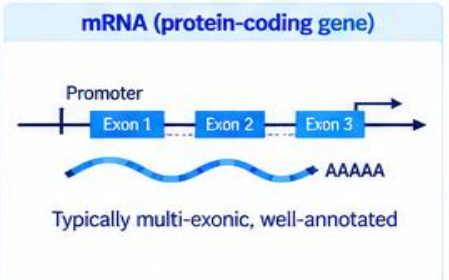
Two RNA worlds, distinct features, shared importance

Not Just
Protein-Coding —
A Bigger RNA Universe

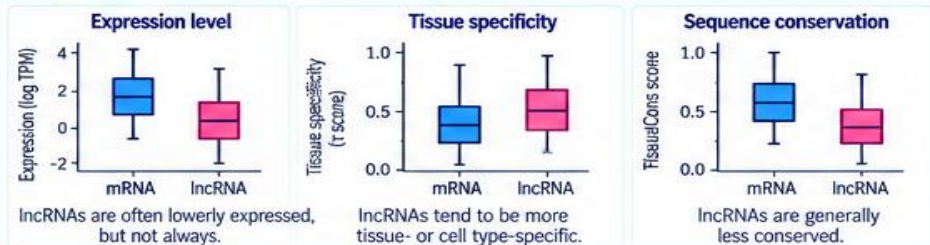
Schematic comparison



Genomic context



Expression characteristics



Feature comparison

Feature	mRNA	lncRNA	Notes
Protein coding	Usually yes	Usually no	lncRNAs have little or no protein-coding potential.
Length	Variable (~0.5–10 kb)	> 200 nt often 0.2–100 kb	lncRNAs are generally longer than 200 nucleotides.
Open reading frame (ORF)	Usually substantial	Usually absent or weak	lncRNAs lack long, conserved ORFs.
Sequence conservation	Often higher	Often lower	lncRNAs are generally less conserved, with some exceptions.
Expression level	Often abundant	Frequently lower	Many lncRNAs are lowly expressed, but not all.
Tissue / cell specificity	Variable	Often high	lncRNAs tend to be more tissue- or cell type-specific.
Annotation status	Relatively mature	Incomplete	lncRNA annotation is continuously improving.
Discovery	Easier (well-established)	More difficult	lncRNAs require specialized pipelines and filtering.
Function	Codes for proteins	Regulatory roles (chromatin, transcription, RNA stability, etc.)	lncRNAs act through diverse mechanisms.
Examples	ACTB, GAPDH, MYC	HOTAIR, MALAT1, NEAT1	Well-studied examples.

Functional outcomes

mRNA

- Translated into protein
- Perform structural, enzymatic and signaling functions
- Essential for many cellular processes
- Often housekeeping or broadly expressed

Proteins execute cellular functions

lncRNA

- Regulate gene expression (at multiple levels)
- Interact with DNA, RNA and proteins
- Modulate chromatin structure
- Influence transcription, RNA stability, translation
- Involved in development, disease and cellular phenotype

lncRNAs regulate gene expression and cellular processes

Key differences at a glance

- mRNA → Protein
- lncRNA → No protein (little or no coding potential)
- lncRNAs are generally longer (> 200 nt)
- lncRNAs are less conserved
- lncRNAs are often lower in expression but more tissue-specific
- mRNA annotation is mature; lncRNA annotation is incomplete
- lncRNAs are harder to discover and function through diverse regulatory mechanisms

Important:
Low expression does not automatically mean lncRNA. Integration of multiple features (e.g., length, ORF, coding potential, conservation) is essential.

Take-home message

- mRNAs and lncRNAs are distinct but complementary components of the transcriptome.
- lncRNAs are not just “noise” — they play important regulatory roles.
- Multiple features must be considered to distinguish lncRNAs from mRNAs.
- Low expression alone is not sufficient to classify lncRNAs.
- Studying lncRNAs provides deeper insights into gene regulation, development and disease.

More Than Coding
More Possibilities

Different RNAs. Diverse Functions. A More Complete View of Biology.



Known
Annotations
Build the Foundation

Known vs Novel lncRNAs

From existing knowledge to new discoveries



New Transcripts
New Insights
New Possibilities

Known lncRNAs

Already present in reference annotation databases.



Reference databases



GENCODE



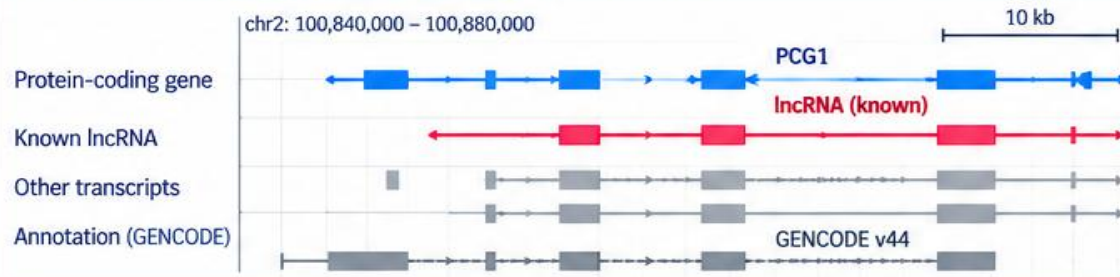
Ensembl



RefSeq

- Curated and well-annotated lncRNAs
- Genomic coordinates, transcript structure and basic information available
- Widely used in downstream analyses (e.g., expression, function, disease association)

Example: Known lncRNA annotation (GENCODE)



Key points

- ✓ Known lncRNAs are obtained from reference annotation databases (e.g., GENCODE, Ensembl, RefSeq).
- ✓ Provide reliable transcript models and functional information (when available).
- ✓ Useful for direct quantification and differential expression analysis.

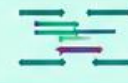
Novel lncRNAs

Not present in the reference annotation and reconstructed from RNA-seq data.

Workflow for identifying novel lncRNAs



RNA-seq reads
(FASTQ files)



Genome alignment
(e.g., HISAT2, STAR)



Transcript assembly
(e.g., StringTie, Scallop)



Novel transcript
(not in reference annotation)



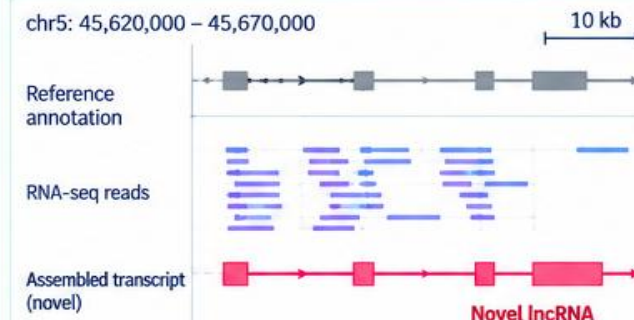
Coding-potential filtering
(e.g., CPC2, CNCI, Pfam, CPAT)



Novel lncRNA
(high-confidence)

Example: Reconstruction of a novel lncRNA

Reference annotation (known)
Novel lncRNA (assembled from RNA-seq)



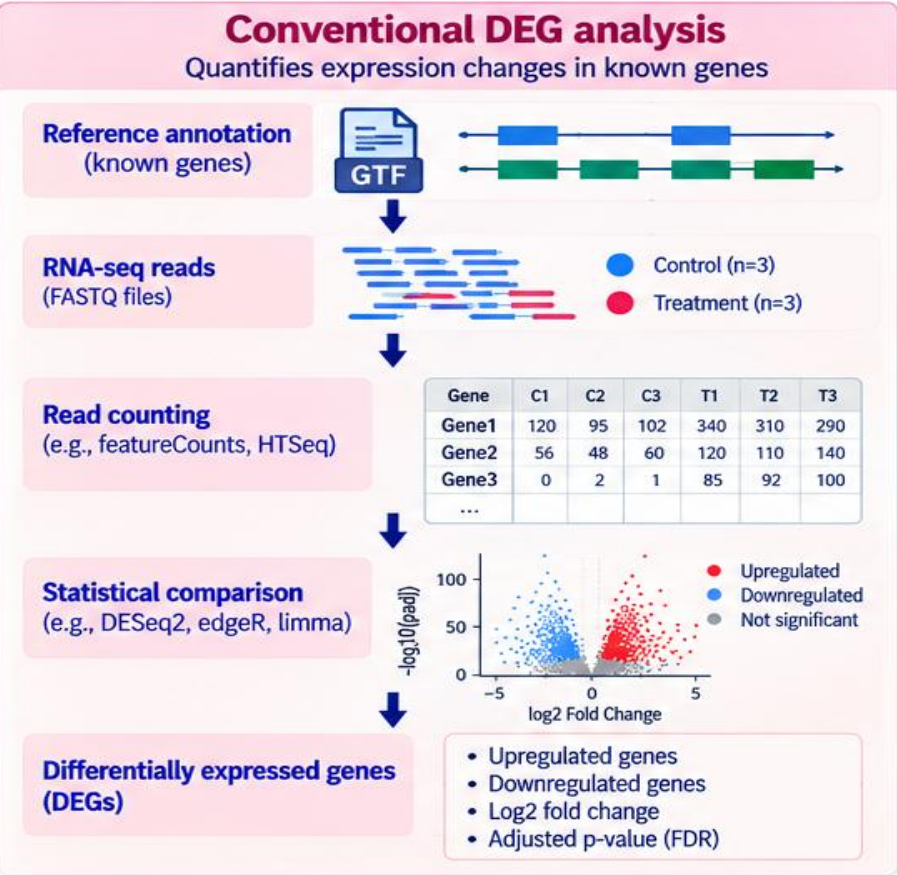
Key points

- ✓ Novel lncRNAs are reconstructed from RNA-seq data.
- ✓ Not present in existing reference annotations.
- ✓ Require stringent filtering to remove potential protein-coding transcripts.
- ✓ Can be tissue-, cell- or condition-specific.
- ✓ Expand the lncRNA catalog and may reveal new biology and disease-associated lncRNAs.

Known Today.
More Tomorrow.

Integrating known and novel lncRNAs for a more complete transcriptome

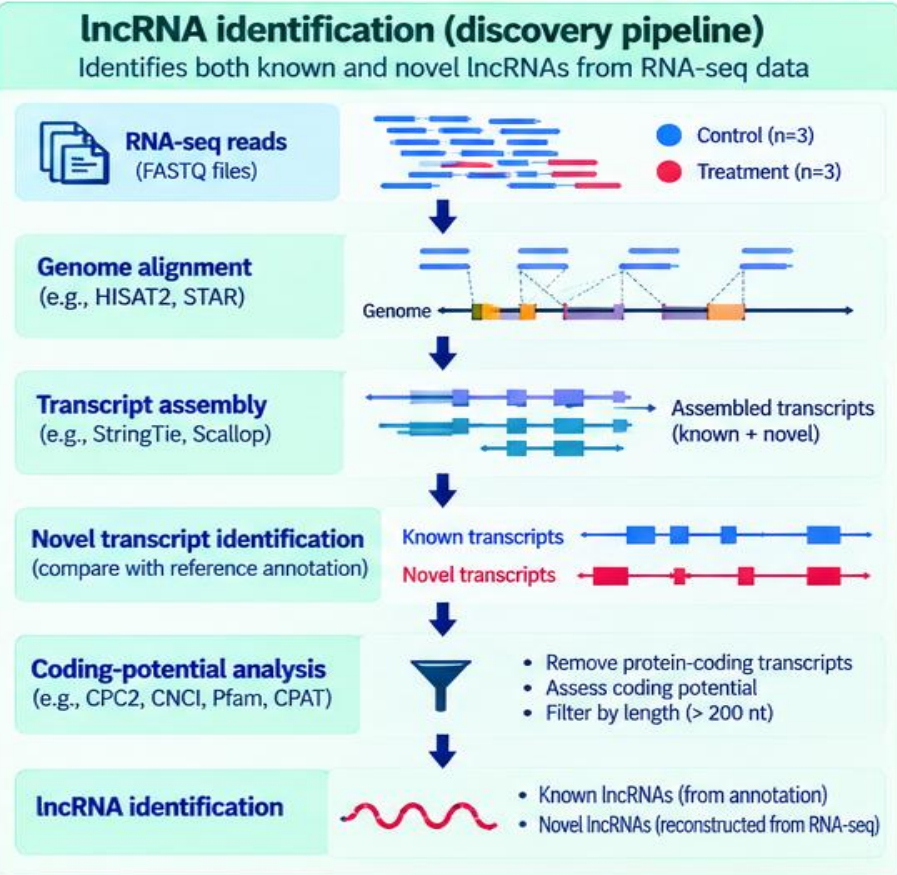
Uncovering the
Unexplored



Key differences		
Feature	DEG analysis	lncRNA identification
Main goal	Identify genes with significant expression changes	Identify and characterize lncRNAs (known and novel)
Input	Known gene annotation (GTF)	Raw RNA-seq reads
Workflow start	Read counting	Genome alignment
Transcript scope	Known genes/transcripts	Known + novel transcripts
Analysis focus	Expression differences between groups	Transcript discovery and coding-potential assessment
Output	List of DEGs (with statistics)	List of lncRNAs (known and novel)
Example tools	featureCounts, DESeq2, edgeR, limma	HISAT2, StringTie, CPC2, CNCI, Pfam, CPAT
Key question	Which known genes are differentially expressed?	What RNA transcripts exist, and which of them have little or no protein-coding potential?

Important points

- DEG analysis and lncRNA identification are not the same analysis.
- Lowly expressed transcripts can be coding or non-coding.
- lncRNA identification requires transcript assembly and coding-potential filtering.
- Both analyses are complementary and can be integrated (e.g., differential expression analysis of known and novel lncRNAs).



DEG analysis asks:

“Which known genes/transcripts are significantly different between groups?”

Focus: changes in expression of annotated protein-coding genes (or other annotated features).

lncRNA discovery asks:

“What RNA transcripts exist, including previously unannotated transcripts, and which of them have little or no protein-coding potential?”

Focus: identification and characterization of known and novel lncRNAs, followed by their quantification and functional analysis.



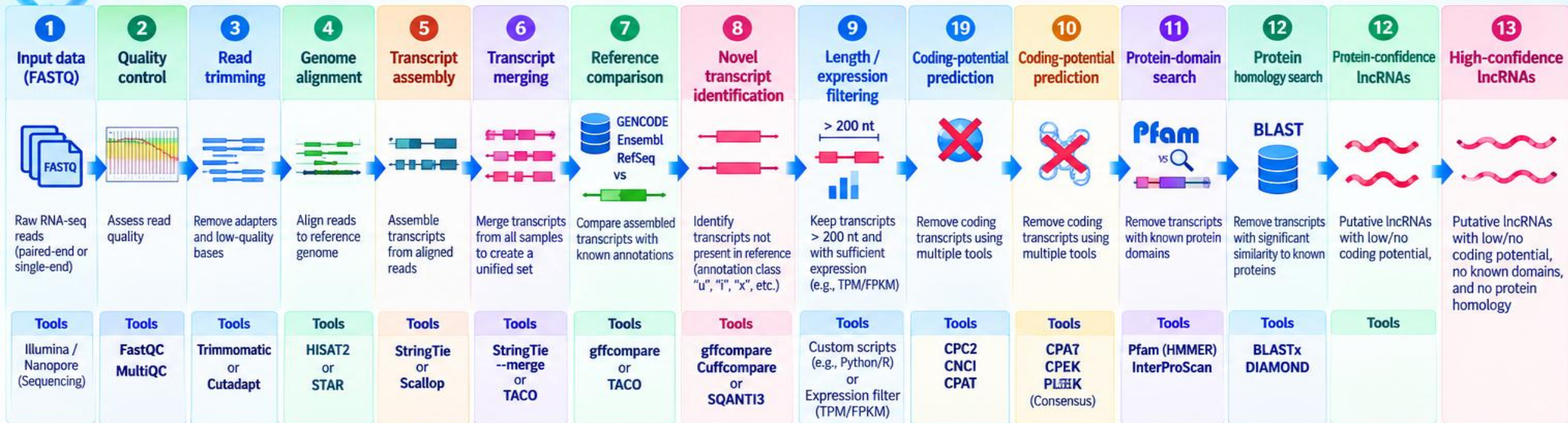
Discover
Annotate
Characterize

lncRNA Discovery Workflow

From RNA-seq Reads to High-Confidence lncRNAs

New Transcripts
New Functions
New Possibilities

Beyond
Protein-Coding
Genes



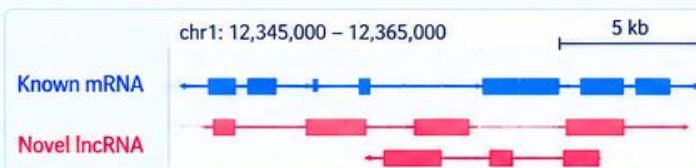
Key filtering criteria

- ✓ Transcript length > 200 nt
- ✓ Minimum expression (e.g., TPM $\geq$ 0.1 or 1)
- ✓ No/weak coding potential (CPC2, CNCI, CPAT, PLEK)
- ✓ No known protein domains (Pfam, InterPro)
- ✓ No significant homology to known proteins (BLASTx/DIAMOND)
- ✓ Classified as novel (not in GENCODE/Ensembl/RefSeq)



Output

- ✓ High-confidence lncRNAs (FASTA/GTF)
- ✓ Classification (intergenic, antisense, intronic, etc.)
- ✓ Basic features (length, expression, genomic location)

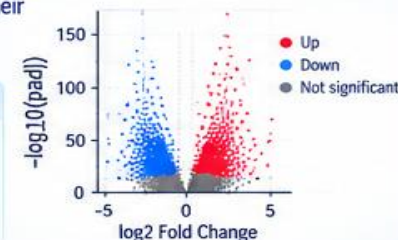


Downstream: Differential expression analysis (of known and novel lncRNAs)

After identifying high-confidence lncRNAs, quantify their expression and perform differential analysis.

Tools

- featureCounts or Salmon/Kallisto (quantification)
- DESeq2 / edgeR / limma (differential expression)



From RNA-seq Reads to Functional Insights
Exploring the Hidden Transcriptome

More than Coding
More to Discover



Different Questions
Different Analyses
Complementary Insights

Two Different Questions in lncRNA Analysis

From Transcript Discovery to Biological Function

First find the lncRNAs
Then study what they do

lncRNAs
Expand the
RNA World

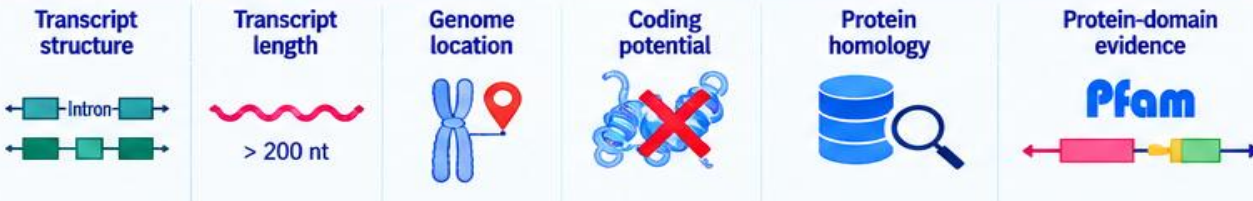
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Discovery: Is this transcript a lncRNA?

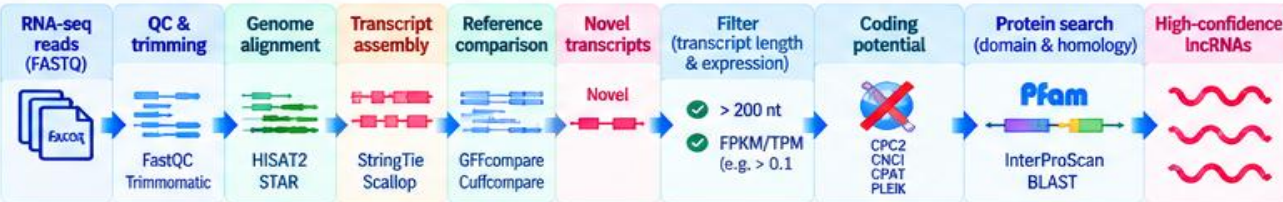
Identify and characterize lncRNAs from RNA-seq and genomic information

What is this transcript?

Required information



Typical workflow for lncRNA identification



Example output

Transcript ID	Length (nt)	Exons	Class	CPC2 score	Pfam hit	BLAST hit	Annotation
MSTRG.1001	1,245	3	Intergenic	-1.23	No	No	Novel lncRNA
MSTRG.2056	856	2	Antisense	-0.98	No	No	Novel lncRNA
XR_001234	2,102	4	Intronic	-1.45	No	No	Novel lncRNA
...	...	...	...	...	...	...	...

lncRNA features

- ✓ Longer than 200 nt
- ✓ Multiple exons
- ✓ Little/no coding potential
- ✓ No significant protein domains
- ✓ No strong homology to known proteins
- ✓ Often tissue- or condition-specific

2

Differential expression: Does this lncRNA change between groups?

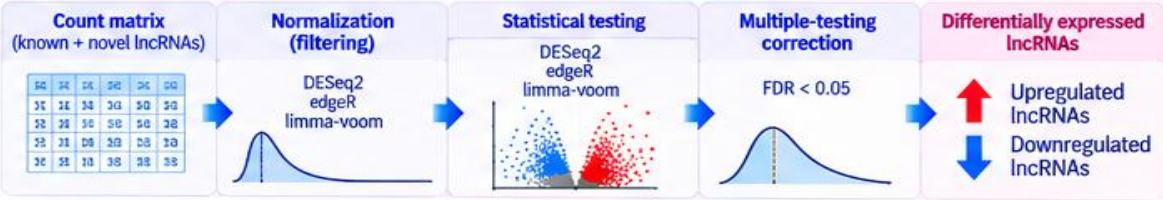
Quantify and test differential expression of identified lncRNAs

Does this lncRNA change?

Required information



Typical workflow for lncRNA differential expression analysis



Example results

lncRNA ID	log2FC	p-value	FDR	Regulation
MSTRG.1001	2.35	1.2e-05	3.4e-04	Up
MSTRG.2056	-1.87	4.5e-04	1.2e-02	Down
XR_001234	3.10	8.1e-06	2.1e-04	Up
...	...	...	...	...

Biological interpretation

- ✓ Identify condition-responsive lncRNAs
- ✓ Link to pathways and functions (e.g., GSEA)
- ✓ Integrate with mRNA, miRNA, epigenomics
- ✓ Validate key lncRNAs (qRT-PCR, functional studies)

Step 1: Identify what is a lncRNA
(Transcript discovery and characterization)

lncRNA identification
comes **before** lncRNA DEG analysis

Step 2: Test whether it changes
(Differential expression analysis)



Discovery builds the catalog
More lncRNAs, more biology



Expression reveals their relevance
From transcripts to function



Together they provide a complete picture
Identify → Quantify → Understand → Validate

Two Questions
One Bigger Understanding



Reference is incomplete
RNA-seq reveals more
Assemble to discover

Why Transcript Assembly is Required?

Essential step for lncRNA discovery from RNA-seq

More Transcripts
More Isoforms
More Biology



1 Reference annotation is incomplete

- Reference GTF/GFF annotations (e.g., GENCODE, Ensembl, RefSeq) do not contain all transcripts.
- Many genes have unannotated isoforms.
- A large number of non-coding transcripts (including lncRNAs) are missing.
- Therefore, relying only on the reference annotation may lead to missed transcripts, especially novel lncRNAs.

Reference annotation



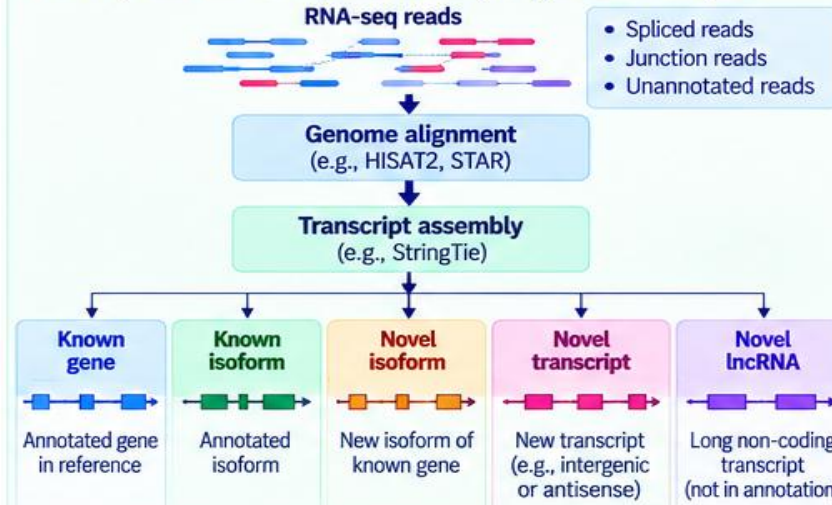
RNA-seq evidence



A reference annotation is not necessarily complete.
RNA-seq can contain transcripts that are absent from the reference.

2 What RNA-seq may detect?

RNA-seq reads can correspond to multiple types of transcripts.



3 Why transcript assembly is needed?

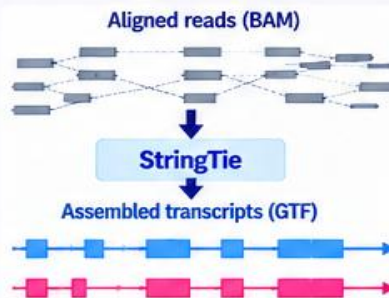
- Reference is incomplete**
 - Many transcripts, isoforms and lncRNAs are not annotated.
- Alignment alone is not enough**
 - Alignment tells where reads map, but not how they are connected into transcripts.
- Discover novel biology**
 - Assembly reconstructs exon-intron structures and transcript boundaries using read evidence.
- Essential for lncRNA discovery**
 - Helps identify novel transcripts and novel lncRNAs.

“ Transcript assembly attempts to reconstruct transcript structures from aligned reads. This is a crucial step when the goal is to discover novel lncRNAs, rather than only quantify already annotated genes.

4 What does StringTie do?

StringTie takes aligned RNA-seq reads (BAM files) and reconstructs transcript models using splice junctions and coverage information.

- ✓ Infers exons and introns
- ✓ Reconstructs transcript boundaries
- ✓ Identifies transcript isoforms
- ✓ Estimates transcript abundance
- ✓ Can identify novel transcripts (not fully present in annotation)



5 StringTie per-sample assembly

Run StringTie separately for each biological sample.

```
stringtie sample.bam \
-G annotation.gtf \
-o sample.gtf \
-p 12
```

Parameter	Meaning
sample.bam	Coordinate-sorted RNA-seq alignment
-G annotation.gtf	Reference annotation (optional but recommended)
-o sample.gtf	Output transcript assembly (GTF)
-p 12	Use 12 CPU threads

- ✓ The reference annotation is supplied with -G, but StringTie can still identify transcript structures that are not completely represented in that annotation.

6 Merge assemblies from all samples

After per-sample assembly, merge all GTF files to obtain a non-redundant set of transcripts.

```
stringtie --merge \
-G annotation.gtf \
-o merged.gtf \
assembly_list.txt
```

Parameter	Meaning
--merge	Merge multiple per-sample assemblies
-G annotation.gtf	Reference annotation
-o merged.gtf	Merged transcript assembly (GTF)
assembly_list.txt	Text file with list of all sample GTF files

★ Key takeaways

- ✓ Transcript assembly is required because reference annotations are incomplete.
- ✓ RNA-seq can detect known genes, known isoforms, novel isoforms, novel transcripts and novel lncRNAs.
- ✓ StringTie reconstructs transcript structures from aligned reads.
- ✓ After merging and filtering, downstream steps (e.g., coding potential analysis) can identify novel lncRNAs.
- ✓ Only after lncRNA identification do we perform differential expression analysis.

Assemble
Discover
Understand

Incomplete
Annotation

Transcript
Assembly

Novel
Transcripts

Novel
lncRNAs

New
Insights

From Reads
to New Biology

1

Purpose

- Compare assembled transcripts with a reference annotation (e.g., GENCODE, Ensembl, RefSeq).
- Classify each transcript based on its genomic relationship to known genes.
- Identify novel transcripts (potential lncRNAs).
- Generate summary statistics and helpful output files for downstream filtering.

2

Command

Basic command:

```
gffcompare \
-r annotation/gencode.gtf \
-o gffcompare/lnc \
merged/merged.gtf
```

Parameter	Meaning
-r annotation/gencode.gtf	Reference annotation (GTF)
-o gffcompare/lnc	Output prefix (creates multiple files)
merged/merged.gtf	Assembled transcripts (from StringTie merge)

3

What gffcompare Produces

Output file	Description
lnc.annotated.gtf	Assembled transcripts with class codes
lnc.transcripts.gtf	Transcript models
lnc.tracking	Mapping between query and reference transcripts
lnc.loci	Loci information
lnc.stats	Summary statistics (class code counts)
lnc.log	Log file

i The annotated GTF file (lnc.annotated.gtf) contains a "class_code" attribute for each transcript, which indicates its relationship to the reference annotation.

4

gffcompare Classification Codes (Class Codes)

Class code	Meaning	Description
=	Exact match	Transcript exactly matches a reference transcript
c	Contained	Transcript is contained within a reference transcript
j	Novel isoform	Potential novel isoform (shares at least one splice junction)
u	Intergenic	Transcript is in an intergenic region (no overlap with known genes)
i	Intronic	Transcript is fully contained within an intron of a known gene
x	Antisense	Transcript overlaps a known gene on the opposite strand
o	Generic exonic overlap	Overlaps a known exon but with different structure
e	Single exon	Single exon transcript overlapping a reference exon
s	Intron match	Overlaps a reference intron
p	Possible polymerase run-on	Potential run-on transcript
r	Repeat	Overlaps a repeat region
n	No overlap	No significant overlap (usually low-confidence)

5

Genomic Relationship of Class Codes

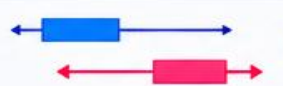
= Exact match  Identical to reference

c Contained  Contained within reference transcript

j Novel isoform  Shares splice junction(s) but different structure

u Intergenic  Located in intergenic region

i Intronic  Inside an intron of a known gene

x Antisense  Overlaps a known gene on the opposite strand

6

Example from lnc.annotated.gtf

```
chr1 StringTie transcript 1000 2500 . + .
gene_id "TCONS_00000123"; transcript_id "TCONS_00000123";
class_code "u";

chr2 StringTie transcript 5000 7200 . - .
gene_id "TCONS_00000456"; transcript_id "TCONS_00000456";
class_code "i";

chr3 StringTie transcript 8000 9100 . + .
gene_id "TCONS_00000789"; transcript_id "TCONS_00000789";
class_code "x";
```

7

Key Points for lncRNA Discovery

- Focus on transcripts with class codes u (intergenic), i (intronic), and x (antisense) as potential novel lncRNAs.
- These transcripts still require further filtering (length, expression, coding potential, protein homology, etc.).
- Other class codes (e.g., j) may also include novel isoforms that could be lncRNAs.

8

Important Note

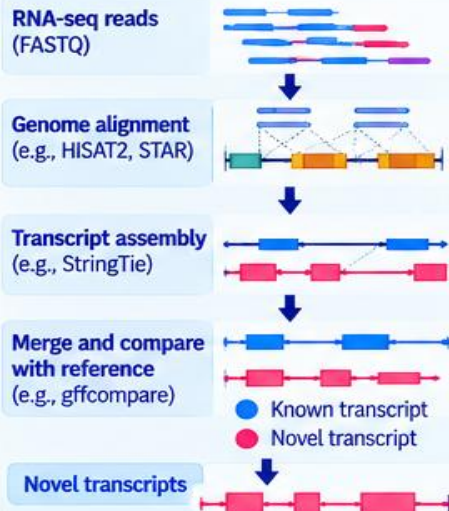
! gffcompare classification only tells you the genomic relationship to the reference annotation. Transcripts classified as u, i, or x are candidate novel lncRNAs, but they must still be filtered for coding potential and other features.

How Do We Prove a Transcript Is Non-Coding?

A stepwise, multi-evidence approach for lncRNA identification

1 Start with a Novel Transcript

Assemble transcripts from RNA-seq and identify novel ones.



2 Filter 1 — Transcript Length

Keep transcripts ≥ 200 nt.



Why?

- ≥ 200 nt is a conventional operational definition to distinguish lncRNAs from many small non-coding RNAs.
- Length alone does not demonstrate non-coding function.
- A long transcript could still encode a protein.

3 Filter 2 — Expression Evidence

Require evidence of genuine transcription.

Transcript X expression (TPM)

Sample	TPM
Control 1	2.4
Control 2	3.1
Control 3	2.7
Tumor 1	5.8
Tumor 2	6.2
Tumor 3	5.1

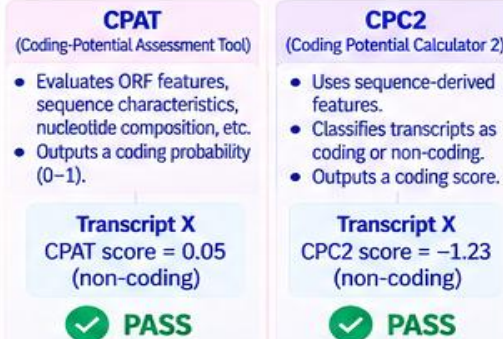
✓ Detectable and reproducible expression → **PASS**

Important:

- Helps remove assembly artifacts from a few isolated reads.
- A common filter is TPM ≥ 1 in a defined number of samples.
- The exact threshold depends on sequencing depth, tissue, study design, and analysis goals.

4 Filter 3 — Coding Potential

Use multiple tools to assess coding capacity.

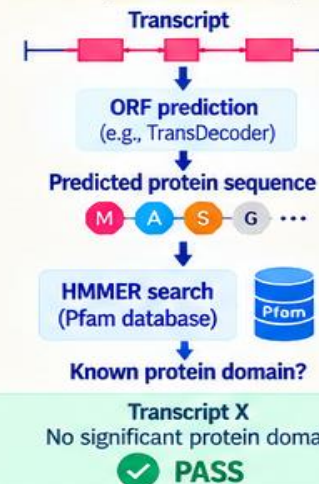


Why use more than one tool?

- Different algorithms use different sequence features.
- Both predict non-coding → stronger evidence.
- If results disagree, the transcript requires further investigation.

5 Filter 4 — Protein-Domain Search

Check for known protein domains in predicted ORF.

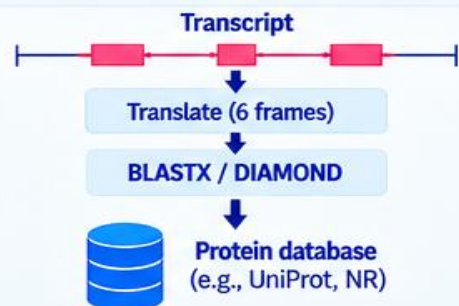


Example (against lncRNA)

✗ Transcript Y: strong kinase domain → likely protein-coding → **REMOVE**

6 Filter 5 — Protein Homology

Search for similarity to known proteins.



Example 1 — No convincing hit

✓ Transcript X
No significant protein similarity
PASS

Example 2 — Strong protein hit

✗ Transcript Y
Strong similarity to known enzyme → likely coding transcript → **REMOVE**

7 Putting All Filters Together

Example: Transcript X (1,245 nt)

- Length ≥ 200 nt **PASS**
- Expression supported **PASS**
- Low coding potential (CPAT + CPC2) **PASS**
- No protein-domain evidence (Pfam/HMMER) **PASS**
- No convincing protein homology (BLASTX/DIAMOND) **PASS**

✓ **Candidate novel lncRNA**

8 Important Scientific Point

! We cannot usually prove absolute non-coding status from RNA-seq alone. Instead, we build a strong computational case that the transcript has no credible protein-coding capacity.

Recommended statement:

"These analyses provide computational evidence that the transcript has low protein-coding potential and lacks significant protein-domain or protein-homology evidence, supporting its classification as a candidate lncRNA."

? Why not absolute proof?

- Some lncRNAs can contain relatively long ORFs.
- Some may have conserved sequence features.
- Some may produce small functional peptides.
- Computational tools are not perfect.
- Experimental validation may be required for specific cases.





Quantify
Compare
Understand

Quantification and Differential Expression Analysis of lncRNAs

From lncRNA Catalogue to Biological Insights

lncRNAs
Also Matter

1 Input: lncRNA Annotation

Use the high-confidence lncRNA set obtained after filtering (novel + known lncRNAs).



lncRNA annotation
(merged/lncRNA.gtf)

```
chr1 StringTie exon 1000 1500 . + . transcript_id "TCONS_000001"; ...
chr1 StringTie exon 2000 2500 . + . transcript_id "TCONS_000001"; ...
chr2 StringTie exon 3000 3200 . - . transcript_id "TCONS_000002"; ...
...
```

2 Read Quantification

Count reads overlapping lncRNA exons using featureCounts (Subread).

```
$ featureCounts \
-T 12 \
-p \
-t exon \
-g transcript_id \
-a merged/lncRNA.gtf \
-o lncRNA_counts.txt \
alignment/*.bam
```

Parameter	Meaning
-T 12	Use 12 CPU threads
-p	Paired-end reads
-t exon	Count reads overlapping exons
-g transcript_id	Use transcript_id as feature ID
-a merged/lncRNA.gtf	lncRNA annotation (GTF)
-o lncRNA_counts.txt	Output count matrix
alignment/*.bam	Input BAM files

3 Count Matrix

featureCounts generates a raw count matrix for all lncRNAs across samples.

Raw count matrix (lncRNAs × samples)

Transcript ID	Control_1	Control_2	Control_3	Tumor_1	Tumor_2	Tumor_3
TCONS_000001	12	18	15	102	95	110
TCONS_000002	0	1	0	44	38	50
TCONS_000003	25	20	22	18	15	17
TCONS_000004	5	4	6	0	1	0
TCONS_000005	100	95	98	120	115	118
...	...	...	...	...	...	...

4 Differential Expression Analysis

Use DESeq2 to identify differentially expressed lncRNAs between groups.



DESeq2 analysis (R)

```
library(DESeq2)
countData <- read.table("lncRNA_counts.txt", header=TRUE, row.names=1)
colData <- read.table("sample_info.txt", header=TRUE, row.names=1)
dds <- DESeqDataSetFromMatrix(countData = countData,
                              colData = colData,
                              design = ~ condition)

dds <- DESeq(dds)
res <- results(dds, contrast = c("condition", "Tumor", "Control"))
resOrdered <- res[order(res$padj), ]
write.csv(as.data.frame(resOrdered), "lncRNA_DESeq2_results.csv")
```

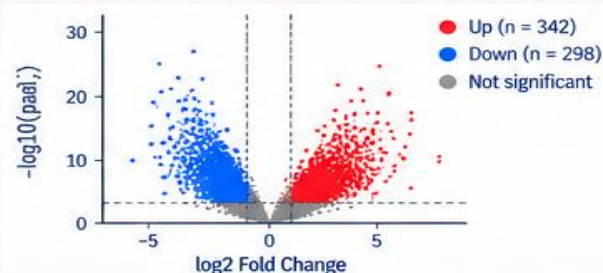
5 DE Results

DESeq2 identifies upregulated and downregulated lncRNAs.

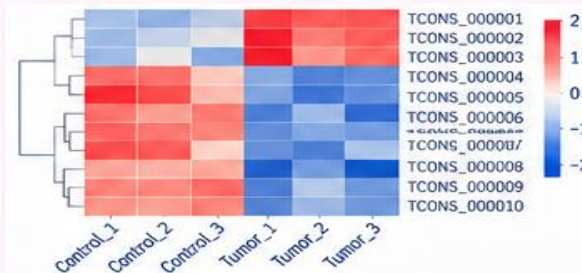
Example result table

Transcript ID	log2FoldChange	pvalue	padj	Regulation
TCONS_000001	3.12	1.2e-05	3.5e-04	Up
TCONS_000002	2.85	4.6e-05	8.1e-04	Up
TCONS_000003	-2.10	9.3e-04	0.021	Down
TCONS_000004	-3.45	1.1e-05	2.9e-04	Down
TCONS_000005	0.32	0.45	0.72	Not significant
...	...	...	...	...

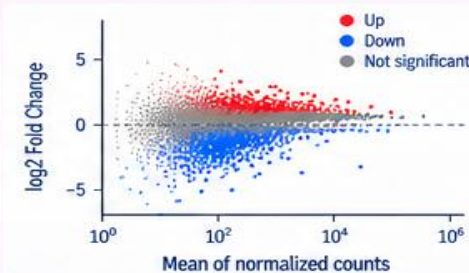
Volcano plot of lncRNA differential expression



Heatmap of differentially expressed lncRNAs



MA plot



6 Downstream Analyses

Differentially expressed lncRNAs can be further analyzed by:



Target prediction
(cis/trans)



Co-expression networks
(WGCNA)



Functional enrichment
(GO/KEGG)



Pathway analysis



Integration with mRNA, miRNA epigenomics, etc.



Key Points

- Quantification is performed after establishing the lncRNA catalogue.
- Use the same alignments as for mRNA, but with lncRNA annotation.
- Raw counts (not TPM/FPKM) are used for DESeq2.
- Include biological replicates and a proper experimental design.
- Apply multiple-testing correction (e.g., FDR < 0.05).
- Visualize results (volcano plot, heatmap, MA plot).



Common Tools

Step	Tools
Read quantification	featureCounts (Subread), HTSeq
Differential expression	DESeq2, edgeR, limma-voom
Visualization	pheatmap, ggplot2, EnhancedVolcano
Functional analysis	clusterProfiler, GSEA
Target prediction	lncTar, miRanda, TargetScan
Network analysis	WGCNA, Cytoscape

lncRNAs
Beyond Protein-Coding Genes

From Quantification to Function
Unlocking the Regulatory Potential of lncRNAs

Small Change
Big Impact

What Can We Do After DE-lncRNA Identification?

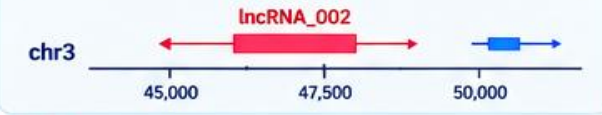
From Expression to Biological Insight

lncRNAs Link Genes, Pathways and Phenotypes

1 Genomic location of lncRNAs

Determine the chromosomal position of each lncRNA.

lncRNA ID	Chromosome	Start	End	Strand
lncRNA_001	chr1	12,345	15,678	+
lncRNA_002	chr3	45,678	48,912	-
lncRNA_003	chr7	102,345	105,987	+
...	...	...	...	...



2 Nearby gene analysis

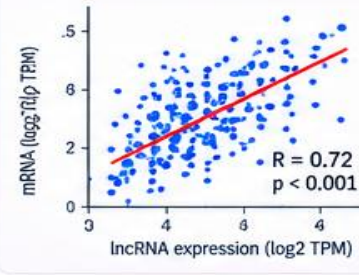
Identify protein-coding genes close to lncRNAs (e.g., within 10–100 kb).



lncRNA ID	Nearest gene	Distance (kb)
lncRNA_001	GeneA	12.4
lncRNA_002	GeneB	35.7
lncRNA_003	GeneC	8.1
...	...	...

3 lncRNA–mRNA correlation

Identify co-expressed lncRNA–mRNA pairs.



Positive or negative correlation may indicate potential regulatory relationships.

4 Target prediction

Predict potential target genes.

Cis targets

- Genes located nearby (e.g., within 10–100 kb)
- May regulate neighboring gene expression



Trans targets

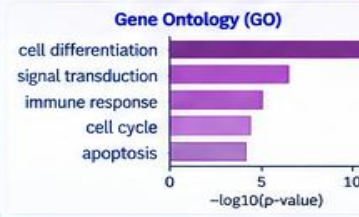
- Genes located distant or on different chromosomes
- Predicted by expression correlation, RNA–RNA interaction, or machine learning



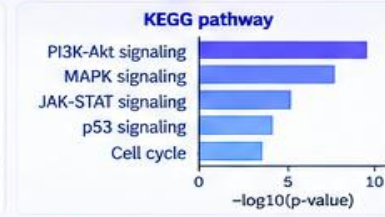
5 Functional enrichment analysis

Use target or co-expressed mRNAs for functional analysis.

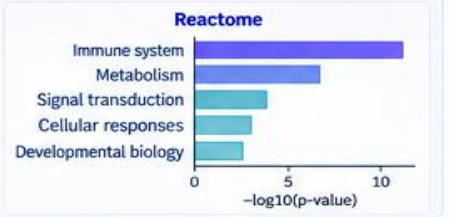
Gene Ontology (GO)



KEGG pathway

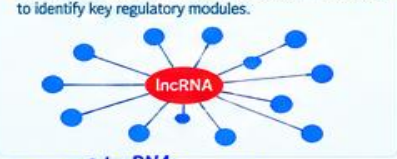


Reactome



6 Network analysis

Construct lncRNA–mRNA co-expression or PPI networks to identify key regulatory modules.



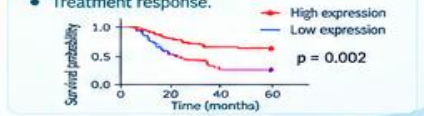
7 Hub lncRNAs

Identify key lncRNAs based on network centrality (e.g., degree, betweenness).

lncRNA ID	Degree	Betweenness	Rank
lncRNA_001	15	0.32	1
lncRNA_002	12	0.28	2
lncRNA_003	10	0.21	3
...	...	...	...

8 Clinical association

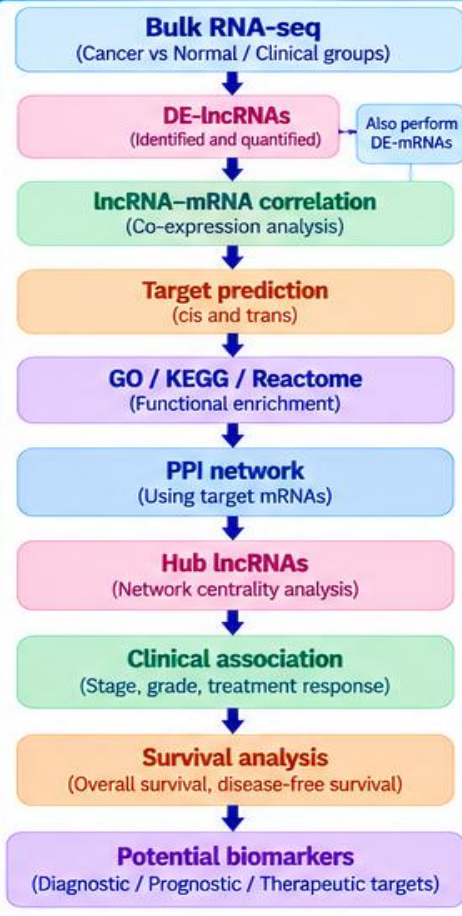
- Association with clinical features.
- Survival analysis (OS, DFS).
- Tumor stage, grade, subtype.
- Treatment response.



Cancer RNA-seq Analysis Pipeline

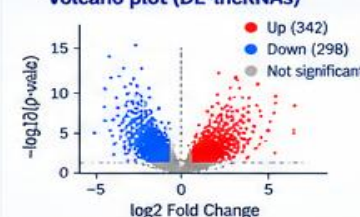
From DE-lncRNAs to Potential Biomarkers

lncRNAs A New Layer in Cancer Biology

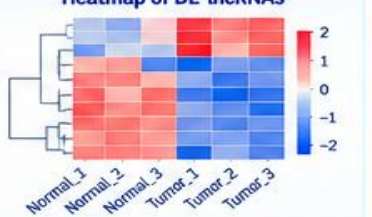


Example analyses

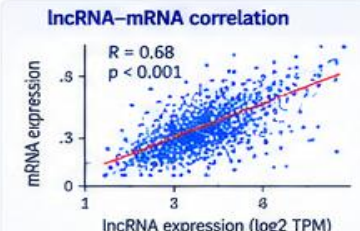
Volcano plot (DE-lncRNAs)



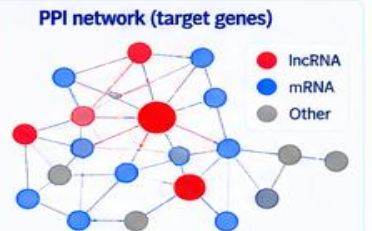
Heatmap of DE-lncRNAs



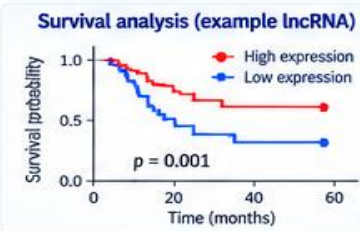
lncRNA–mRNA correlation



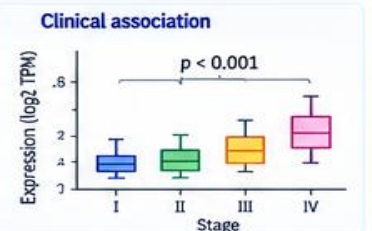
PPI network (target genes)



Survival analysis (example lncRNA)



Clinical association



Other possible analyses

- Disease-free survival (DFS)
- Tumor vs normal comparison
- Subtype-specific lncRNAs
- lncRNAs associated with drug response
- Integration with other omics (miRNA, methylation)
- Construction of ceRNA networks
- Pan-cancer analysis
- Experimental validation (qRT-PCR)

Final Take-Home Message

Key Concepts and Complete Workflow

Three separate concepts

- lncRNA discovery**
What transcripts are present and which are non-coding?
- lncRNA annotation**
Which are known and which are novel?
- Differential expression**
Which identified lncRNAs change significantly between experimental groups?

Complete workflow

- RNA-seq → QC → Alignment (STAR)
- Transcript assembly (StringTie)
- Reference comparison (gffcompare)
- Known and novel transcripts
- Coding-potential filtering (CPAT + CPC2)
- Protein-domain and homology search (Pfam + BLAST)
- High-confidence lncRNAs
- Quantification (featureCounts)
- Differential expression (DESeq2)
- Functional and clinical analysis

Key distinction

- DEG analysis** = expression comparison (known genes)
- lncRNA identification** = transcript discovery + non-coding validation
- DE-lncRNA analysis** = lncRNA identification + expression comparison

“Do not simply extract lncRNAs from a standard DEG table. lncRNA discovery requires transcript assembly and non-coding validation.”

21 lncRNA Annotation: Databases Used

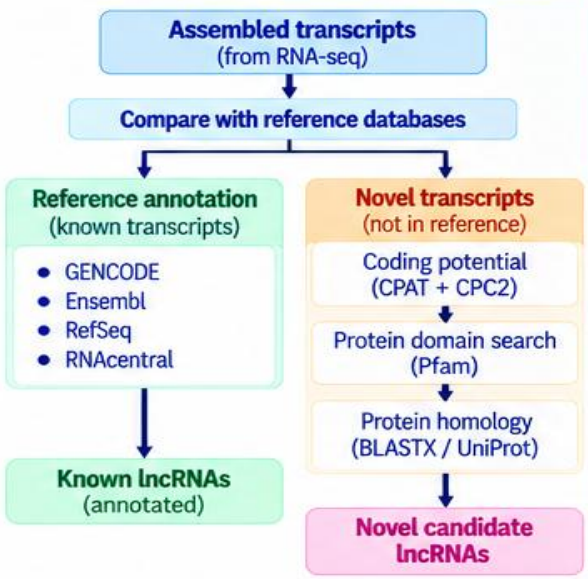
Multiple databases provide complementary information

Purpose	Database	What it tells us
Known lncRNAs	GENCODE	Known human lncRNA genes/transcripts
Alternative annotation	Ensembl	Gene/transcript annotation
RefSeq annotation	NCBI RefSeq	Curated transcripts
Other ncRNAs	RNAcentral	Integrated ncRNA annotation
miRNA	miRBase	Known miRNAs
snoRNA/snrRNA	snoDB / RNAcentral	Small ncRNAs
rRNA	SILVA	Ribosomal RNA
Protein homology	UniProt	Known proteins
Protein domains	Pfam	Conserved protein domains
Coding potential	CPAT / CPC2	Protein-coding potential
Repeats	RepeatMasker/Repbse	Repetitive elements

Use multiple databases to obtain a comprehensive and accurate annotation of lncRNAs and to remove other RNA classes.

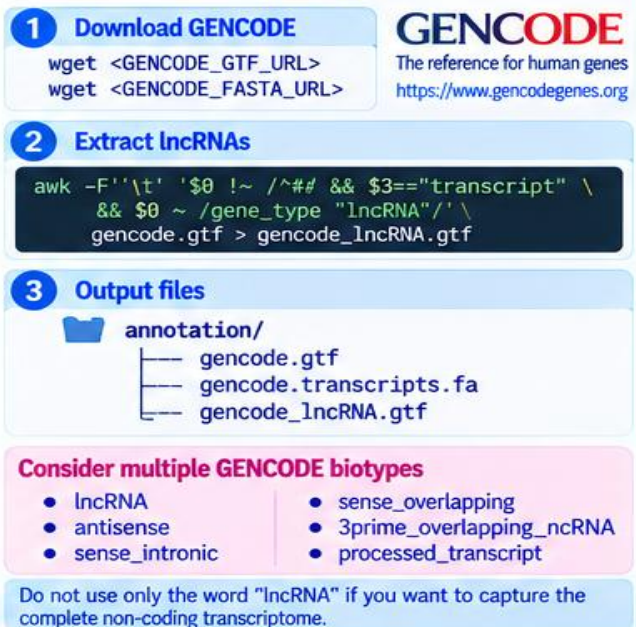
22 Recommended Annotation Hierarchy

Combine multiple databases for robust annotation



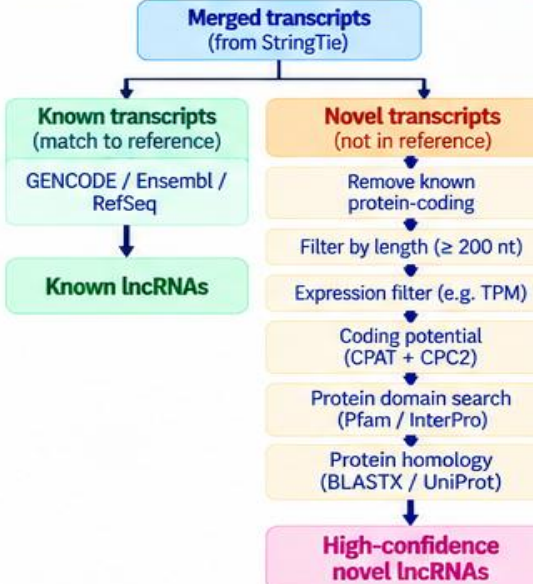
23 Known lncRNA Annotation with GENCODE

GENCODE is the primary resource for human lncRNAs



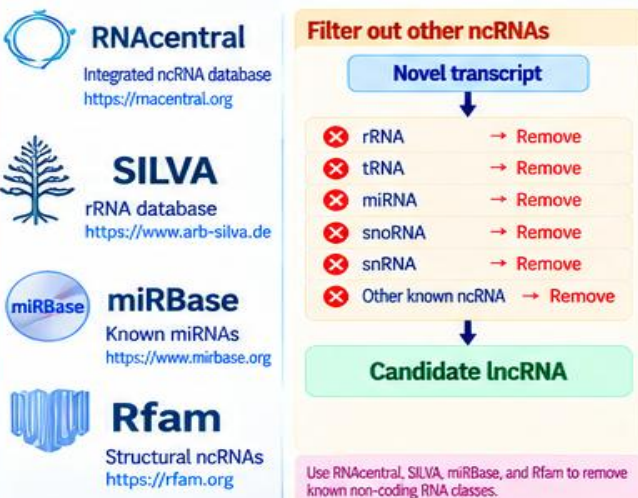
24 Annotation of Novel Transcripts

Identify high-confidence novel lncRNAs



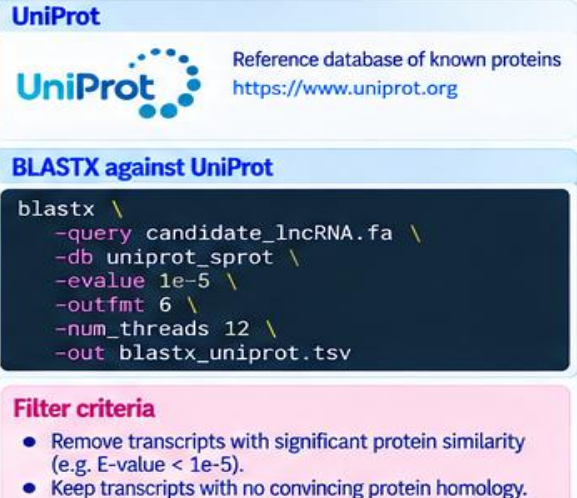
25 Remove Other Non-Coding RNAs

Not every non-protein-coding transcript is a lncRNA



26 Protein-Coding Contamination Check

Remove transcripts with similarity to known proteins



27 Protein Domain Annotation

Identify conserved protein domains



28 Recommended Final Annotation Strategy

Complete pipeline for human lung cancer RNA-seq

